Assignment 3 Fundamentals ML

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R Markdown

Librarys

```
library(reshape2)
```

```
## Warning: package 'reshape2' was built under R version 4.5.2
```

```
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.5.2
```

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   4.0.0     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ ggplot2::element() masks e1071::element()
## ✖ dplyr::filter()    masks stats::filter()
## ✖ dplyr::lag()       masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(class)
```

read Universal Bank file

```
UVBank <- read.csv("C:/Users/ptobo/Desktop/Kent State/3 Fundamentals of Machine Learning BA-6406
0/Assignments/Assignment 2/UniversalBank.csv")
```

rename personal loan to loan and credit card to cc, then convert to categorical factor

```
summary(UVBank)
```

```
##        ID            Age          Experience       Income          ZIP.Code
##  Min.   :   1   Min.   :23.00   Min.   :-3.0   Min.   :  8.00   Min.   : 9307
##  1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
##  Median :2500   Median :45.00   Median :20.0   Median : 64.00   Median :93437
##  Mean   :2500   Mean   :45.34   Mean   :20.1   Mean   : 73.77   Mean   :93153
##  3rd Qu.:3750   3rd Qu.:55.00   3rd Qu.:30.0   3rd Qu.: 98.00   3rd Qu.:94608
##  Max.   :5000   Max.   :67.00   Max.   :43.0   Max.   :224.00   Max.   :96651
##      Family          CCAvg          Education        Mortgage
##  Min.   :1.000   Min.   : 0.000   Min.   :1.000   Min.   :  0.0
##  1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.000   1st Qu.:  0.0
##  Median :2.000   Median : 1.500   Median :2.000   Median :  0.0
##  Mean   :2.396   Mean   : 1.938   Mean   :1.881   Mean   : 56.5
##  3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.000   3rd Qu.:101.0
##  Max.   :4.000   Max.   :10.000   Max.   :3.000   Max.   :635.0
##  Personal.Loan   Securities.Account   CD.Account         Online
##  Min.   :0.000   Min.   :0.0000     Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.000   1st Qu.:0.0000     1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.000   Median :0.0000     Median :0.0000   Median :1.0000
##  Mean   :0.096   Mean   :0.1044     Mean   :0.0604   Mean   :0.5968
##  3rd Qu.:0.000   3rd Qu.:0.0000     3rd Qu.:0.0000   3rd Qu.:1.0000
##  Max.   :1.000   Max.   :1.0000     Max.   :1.0000   Max.   :1.0000
##    CreditCard
##  Min.   :0.000
##  1st Qu.:0.000
##  Median :0.000
##  Mean   :0.294
##  3rd Qu.:1.000
##  Max.   :1.000
```

```
names(UVBank)[names(UVBank)=="CreditCard"] <- "CC"
names(UVBank)[names(UVBank)=="Personal.Loan"] <- "Loan"
summary(UVBank)
```

```
##        ID            Age          Experience       Income          ZIP.Code
##  Min.   :   1   Min.   :23.00   Min.   :-3.0   Min.   :  8.00   Min.   : 9307
##  1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
##  Median :2500   Median :45.00   Median :20.0   Median : 64.00   Median :93437
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##  Max.   :5000   Max.   :67.00   Max.   :43.0   Max.   :224.00   Max.   :96651
##      Family          CCAvg          Education        Mortgage
##  Min.   :1.000   Min.   : 0.000   Min.   :1.000   Min.   :  0.0
##  1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.000   1st Qu.:  0.0
##  Median :2.000   Median : 1.500   Median :2.000   Median :  0.0
##  Mean   :2.396   Mean   : 1.938   Mean   :1.881   Mean   : 56.5
##  3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.000   3rd Qu.:101.0
##  Max.   :4.000   Max.   :10.000   Max.   :3.000   Max.   :635.0
##       Loan       Securities.Account   CD.Account         Online
##  Min.   :0.000   Min.   :0.0000     Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.000   1st Qu.:0.0000     1st Qu.:0.0000   1st Qu.:0.0000
##  Median :0.000   Median :0.0000     Median :0.0000   Median :1.0000
##  Mean   :0.096   Mean   :0.1044     Mean   :0.0604   Mean   :0.5968
##  3rd Qu.:0.000   3rd Qu.:0.0000     3rd Qu.:0.0000   3rd Qu.:1.0000
##  Max.   :1.000   Max.   :1.0000     Max.   :1.0000   Max.   :1.0000
##        CC
##  Min.   :0.000
##  1st Qu.:0.000
##  Median :0.000
##  Mean   :0.294
##  3rd Qu.:1.000
##  Max.   :1.000
```

```
UVBank$CC <- factor(UVBank$CC, levels = c(0,1))
UVBank$Loan <- factor(UVBank$Loan, levels = c(0,1))
UVBank$Online <- factor(UVBank$Online, levels = c(0,1))

summary(UVBank)
```

```
##        ID            Age          Experience       Income          ZIP.Code
##  Min.   :   1   Min.   :23.00   Min.   :-3.0   Min.   :  8.00   Min.   : 9307
##  1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
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##  Max.   :5000   Max.   :67.00   Max.   :43.0   Max.   :224.00   Max.   :96651
##      Family          CCAvg          Education        Mortgage     Loan
##  Min.   :1.000   Min.   : 0.000   Min.   :1.000   Min.   :  0.0   0:4520
##  1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.000   1st Qu.:  0.0   1: 480
##  Median :2.000   Median : 1.500   Median :2.000   Median :  0.0
##  Mean   :2.396   Mean   : 1.938   Mean   :1.881   Mean   : 56.5
##  3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.000   3rd Qu.:101.0
##  Max.   :4.000   Max.   :10.000   Max.   :3.000   Max.   :635.0
##  Securities.Account   CD.Account     Online   CC
##  Min.   :0.0000     Min.   :0.0000   0:2016   0:3530
##  1st Qu.:0.0000     1st Qu.:0.0000   1:2984   1:1470
##  Median :0.0000     Median :0.0000
##  Mean   :0.1044     Mean   :0.0604
##  3rd Qu.:0.0000     3rd Qu.:0.0000
##  Max.   :1.0000     Max.   :1.0000
```

set seed to 123

```
set.seed(123)
```

train 60/40

```
train60 <- sample(nrow(UVBank),.6 * nrow(UVBank))
train <- UVBank[train60, ]
valid <- UVBank[-train60, ]

summary(UVBank)
```

```
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##  Min.   :   1   Min.   :23.00   Min.   :-3.0   Min.   :  8.00   Min.   : 9307
##  1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
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##  3rd Qu.:3750   3rd Qu.:55.00   3rd Qu.:30.0   3rd Qu.: 98.00   3rd Qu.:94608
##  Max.   :5000   Max.   :67.00   Max.   :43.0   Max.   :224.00   Max.   :96651
##      Family          CCAvg          Education        Mortgage     Loan
##  Min.   :1.000   Min.   : 0.000   Min.   :1.000   Min.   :  0.0   0:4520
##  1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.000   1st Qu.:  0.0   1: 480
##  Median :2.000   Median : 1.500   Median :2.000   Median :  0.0
##  Mean   :2.396   Mean   : 1.938   Mean   :1.881   Mean   : 56.5
##  3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.000   3rd Qu.:101.0
##  Max.   :4.000   Max.   :10.000   Max.   :3.000   Max.   :635.0
##  Securities.Account   CD.Account     Online   CC
##  Min.   :0.0000     Min.   :0.0000   0:2016   0:3530
##  1st Qu.:0.0000     1st Qu.:0.0000   1:2984   1:1470
##  Median :0.0000     Median :0.0000
##  Mean   :0.1044     Mean   :0.0604
##  3rd Qu.:0.0000     3rd Qu.:0.0000
##  Max.   :1.0000     Max.   :1.0000
```

```
cat("trainedRows:", nrow(train), "validRows", nrow(valid))
```

```
## trainedRows: 3000 validRows 2000
```

melted pivot tables

```
melted <- melt(train, id.vars = c("CC", "Loan", "Online"), measure.vars = "ID")
pivot <-  dcast(melted, CC + Loan ~ Online, fun.aggregate = length)

pivot
```

```
##   CC Loan   0    1
## 1  0    0 785 1145
## 2  0    1  65  122
## 3  1    0 317  475
## 4  1    1  34   57
```

group by category, CC w/loan

```
x3 <- pivot[pivot$CC == 1 & pivot$Loan == 1, "1"]
x3
```

```
## [1] 57
```

CC, no Loan

```
x2 <- pivot[pivot$CC == 1 & pivot$Loan == 0, "1"]
x2
```

```
## [1] 475
```

pivot table probability

```
prob_x <- x3 / (x3 + x2)
prob_x
```

```
## [1] 0.1071429
```

naive bayes catagories

```
Loan_Online <-  table(Loan = train$Loan, Online = train$Online)
Loan_CC <- table(Loan = train$Loan, CC = train$CC)

Loan_Online
```

```
##     Online
## Loan    0    1
##    0 1102 1620
##    1   99  179
```

```
Loan_CC
```

```
##     CC
## Loan    0    1
##    0 1930  792
##    1  187   91
```

```
Loan_Online
```

```
##     Online
## Loan    0    1
##    0 1102 1620
##    1   99  179
```

```
Loan_CC
```

```
##     CC
## Loan    0    1
##    0 1930  792
##    1  187   91
```

Compute the following quantities means “the probability of A given B”

```
p_i <- Loan_CC["1","1"] / sum(Loan_CC["1",])
p_i
```

```
## [1] 0.3273381
```

```
p_ii <- Loan_Online["1", "1"] / sum(Loan_Online["1", ])
p_ii
```

```
## [1] 0.6438849
```

```
p_iii <- mean(train$Loan == 1)
p_iii
```

```
## [1] 0.09266667
```

```
p_iv <- Loan_CC["0", "1"] / sum(Loan_CC["0", ])
p_iv
```

```
## [1] 0.2909625
```

```
p_v <- Loan_Online["0", "1"] / sum(Loan_Online["0", ])
p_v
```

```
## [1] 0.5951506
```

```
p_vi <- mean(train$Loan == 0)

p_vi
```

```
## [1] 0.9073333
```

Naive Bayes (NB) probability

```
Numerator <- p_i * p_ii * p_iii
Denominator <-  Numerator + p_iv * p_v * p_vi
P_NB <- Numerator/Denominator
P_NB
```

```
## [1] 0.1105637
```

```
NB <- naiveBayes(Loan ~ CC + Online, data = train)

NB
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##          0          1
## 0.90733333 0.09266667
##
## Conditional probabilities:
##    CC
## Y           0         1
##   0 0.7090375 0.2909625
##   1 0.6726619 0.3273381
##
##    Online
## Y           0         1
##   0 0.4048494 0.5951506
##   1 0.3561151 0.6438849
```

new customer

```
New_Customer <- data.frame(CC = factor(1, levels = c(0,1)), Online = factor(1, levels = c(0,
1)))
New_Customer
```

```
##   CC Online
## 1  1      1
```

```
predict(NB, New_Customer, type = "raw")[, "1"]
```

```
##         1
## 0.1105637
```

Based on comparison of the Pivot table results; (prob_x <- x3 / (x3 + x2) = 0.1071429) vs (predict(NB, New_Customer, type = “raw”)[, “1”]) = 0.1105637, Pivot probability is more accurate based on zero assumptions with smaller datasets, whereas NB uses small biased assumptions used to identify patterns resulting in a slightly higher prediction value that may be considered negligible.